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Artificial Intelligence Section

PRACTICAL NO 1

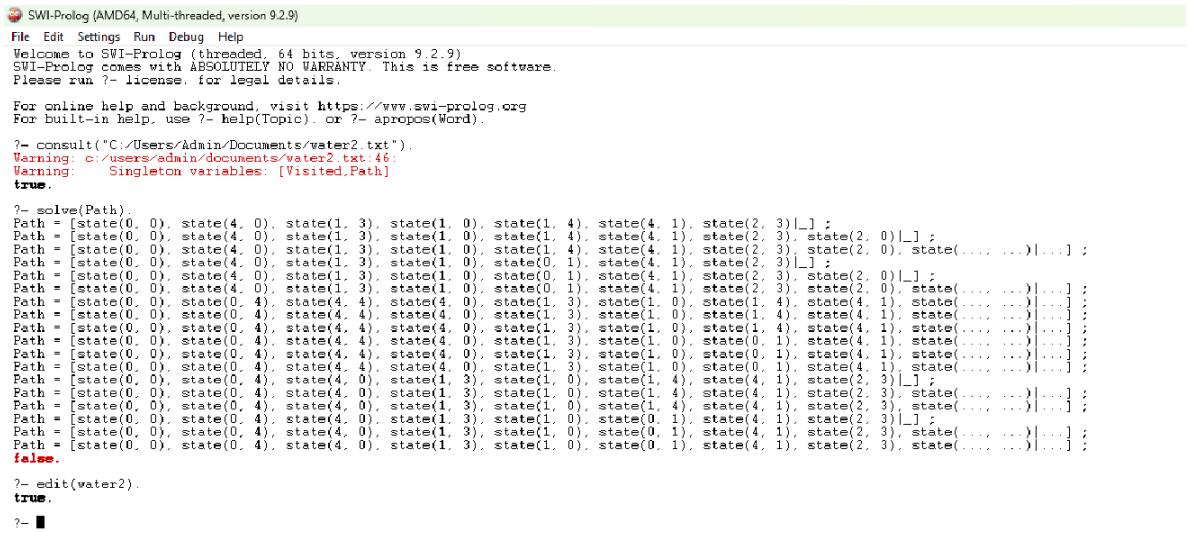
Implementation of Logic programming using PROLOG DFS for water jug problem

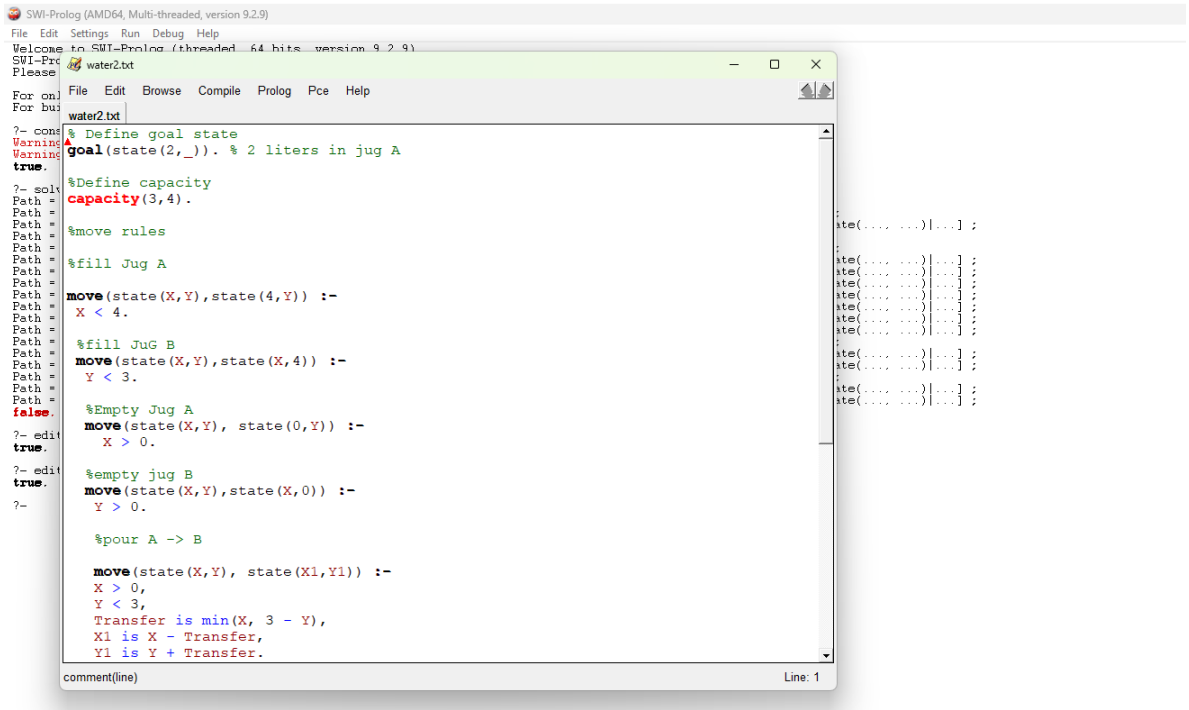
CODE:

```
/* Water Jug Problem using Prolog */

/* Goal state: 2 liters in Jug A */
goal(state(2,_)).

/* Capacities of jugs */
capacity(4,3).

/* Move rules */

/* Fill Jug A */
move(state(X,Y), state(4,Y)) :-
   X < 4.

/* Fill Jug B */
move(state(X,Y), state(X,3)) :-
   Y < 3.

/* Empty Jug A */
move(state(X,Y), state(0,Y)) :-
   X > 0.

/* Empty Jug B */
move(state(X,Y), state(X,0)) :-
   Y > 0.

/* Pour A -> B */
move(state(X,Y), state(X1,Y1)) :-
   X > 0,
   Y < 3,
   Transfer is min(X, 3-Y),
   X1 is X - Transfer,
   Y1 is Y + Transfer.

/* Pour B -> A */
move(state(X,Y), state(X1,Y1)) :-
   Y > 0,
   X < 4,
   Transfer is min(Y, 4-X),
   Y1 is Y - Transfer,
   X1 is X + Transfer.

/* Path search */
path(State, _) :-
   goal(State),
   write('Goal reached: '), write(State), nl.
```

```
path(State, Visited) :-
    move(State, NextState),
    \+ member(NextState, Visited),
    write(NextState), nl,
    path(NextState, [NextState|Visited]).

/* Start */
waterjug :-
    path(state(0,0), [state(0,0)]).
```

OUTPUT:





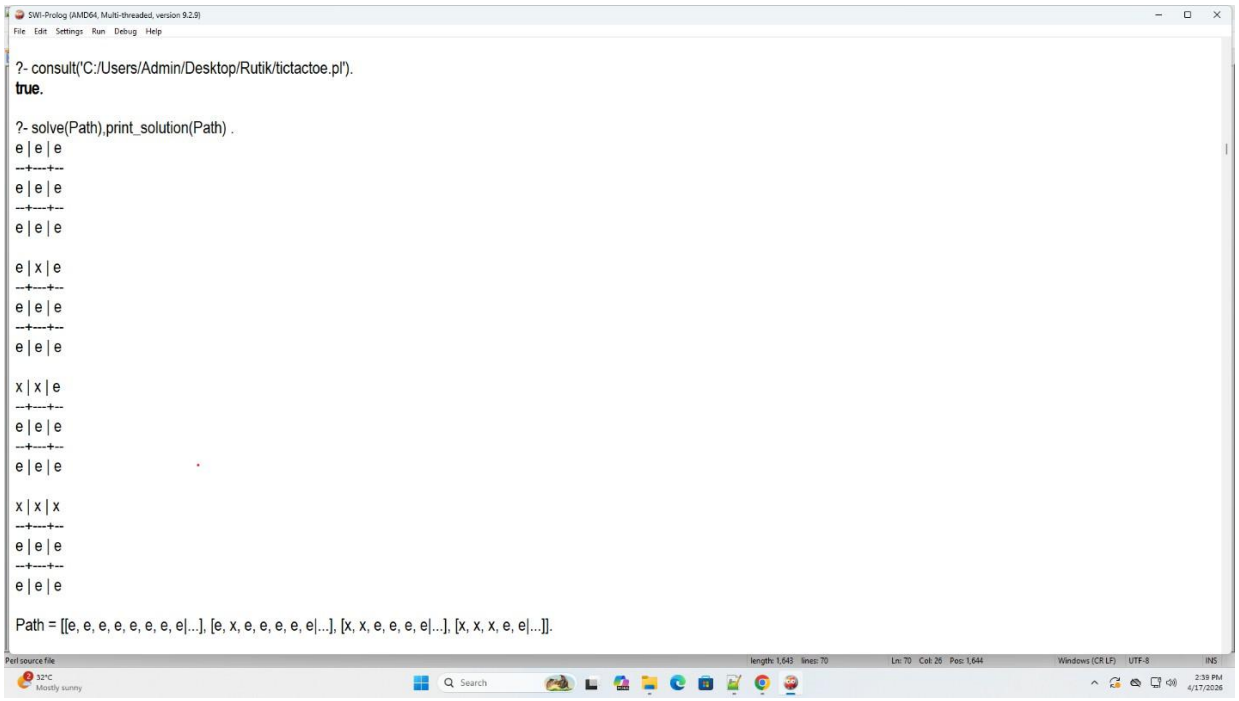
PRACTICAL NO 2
Implementation of Logic programming using PROLOG BFS for tic-tac-toe problem

CODE:

```
% --- Tic Tac Toe BFS Solver ---

% Initial empty board
initial([e,e,e,
         e,e,e,
         e,e,e]).

% Goal: X wins
goal(Board) :- win(Board, x).

% Winning conditions
win([P,P,P,_,_,_,_,_,_],P).
win([_,_,_,P,P,P,_,_,_],P).
win([_,_,_,_,_,_,P,P,P],P).
win([P,_,_,P,_,_,P,_,_],P).
win([_,P,_,_,P,_,_,P,_],P).
win([_,_,P,_,_,P,_,_,P],P).
win([P,_,_,_,P,_,_,_,P],P).
win([_,_,P,_,P,_,P,_,_],P).

% Switch player
next_player(x, o).
next_player(o, x).

% Generate next moves
move(Board, Player, NewBoard) :-
    nth0(Index, Board, e),          % find empty cell
    replace(Board, Index, Player, NewBoard).

% Replace element in list
replace([_|T], 0, X, [X|T]).
replace([H|T], I, X, [H|R]) :-
    I > 0,
    I1 is I - 1,
    replace(T, I1, X, R).

% --- BFS Implementation ---

bfs([[Board|Path]|_], _, [Board|Path]) :-
    goal(Board), !.

bfs([[Board|Path]|Queue], Player, Solution) :-
    findall(
        [NextBoard,Board|Path],
        move(Board, Player, NextBoard),
        Children
    ),
    append(Queue, Children, NewQueue),
```

```
    next_player(Player, NextPlayer),
    bfs(NewQueue, NextPlayer, Solution).

% Solve predicate
solve(Path) :-
    initial(Board),
    bfs([[Board]], x, RevPath),
    reverse(RevPath, Path).

% Print board
print_board([A,B,C,D,E,F,G,H,I]) :-
    format("~w | ~w | ~w~n",[A,B,C]),
    format("--+---+--~n"),
    format("~w | ~w | ~w~n",[D,E,F]),
    format("--+---+--~n"),
    format("~w | ~w | ~w~n~n",[G,H,I]).

% Print solution path
print_solution([]).
print_solution([Board|Rest]) :-
    print_board(Board),
    print_solution(Rest).
```

OUTPUT-



PRACTICAL NO 3

Implementation of Logic programming using PROLOG Hill-climbing to solve 8- Puzzle Problem.

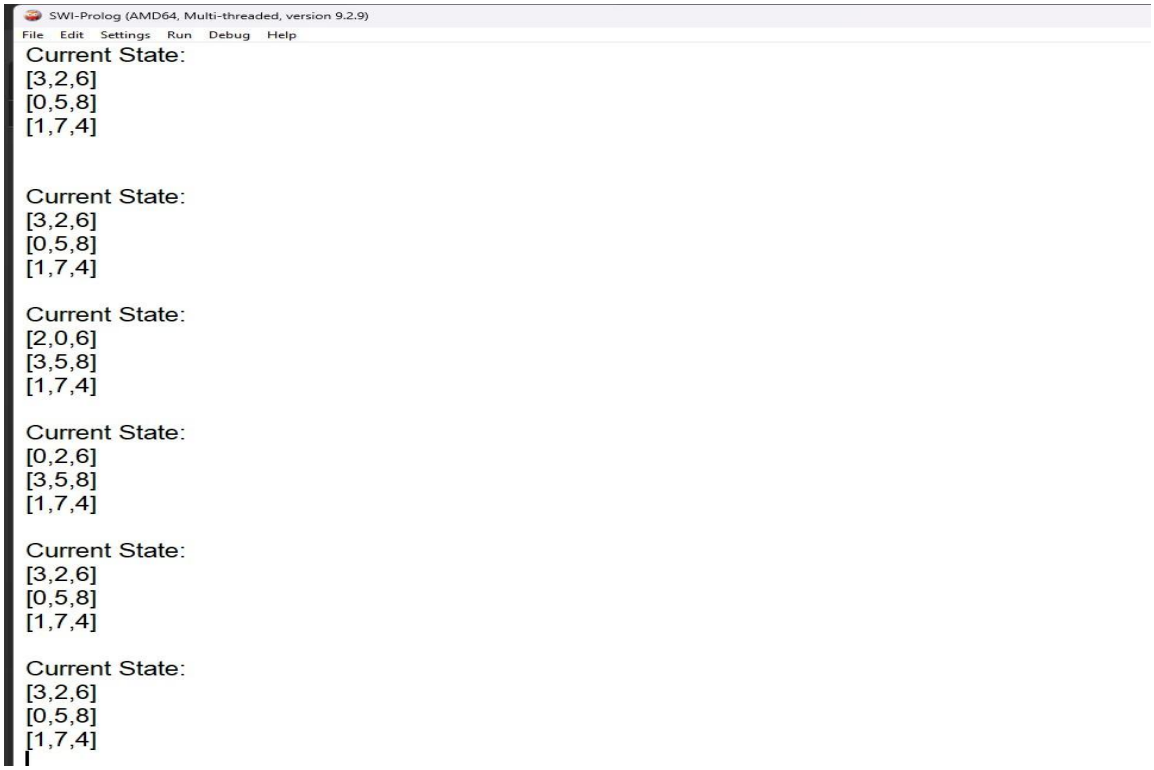
CODE:

```
% Goal State (as a flat list)
goal([1,2,3,4,5,6,7,8,0]).  % 0 represents blank

% Entry point
solve(Start) :-
    hill_climb(Start, []).

% Hill Climbing Algorithm
hill_climb(State, _) :-
    goal(State),
    write('Goal Reached:'), nl,
    print_state(State).

hill_climb(State, Visited) :-
    get_children(State, Children),
    exclude(member_of(Visited), Children, ValidChildren),
    evaluate_children(ValidChildren, ScoredChildren),
    sort(ScoredChildren, Sorted),   % sort by heuristic
    Sorted = [[_, BestChild] | _],  % pick best child
    write('Current State:'), nl,
    print_state(State), nl,
    hill_climb(BestChild, [State | Visited]).

% Check membership
member_of(List, Element) :-
    member(Element, List).

% Generate children (possible moves)
get_children(State, Children) :-
    findall(Child, move(State, Child), Children).

% Possible Moves
move(State, NewState) :-
    swap(State, 0, -3, NewState). % up
move(State, NewState) :-
    swap(State, 0, 3, NewState).  % down
move(State, NewState) :-
    swap(State, 0, -1, NewState). % left
move(State, NewState) :-
    swap(State, 0, 1, NewState).  % right

% Swap positions
swap(State, Zero, Move, NewState) :-
    nth0(Index, State, Zero),
    NewIndex is Index + Move,

    valid_move(Index, NewIndex),
```

```
    nth0(NewIndex, State, Value),
    set_element(State, Index, Value, Temp),

    set_element(Temp, NewIndex, Zero, NewState).

% Valid moves
valid_move(I, J) :-
    J >= 0, J < 9,
    \+ invalid_edge(I, J).

invalid_edge(2, 3).
invalid_edge(3, 2).
invalid_edge(5, 6).
invalid_edge(6, 5).

% Replace element in list
set_element([_|T], 0, X, [X|T]).
set_element([H|T], I, X, [H|R]) :-
    I > 0,
    I1 is I - 1,
    set_element(T, I1, X, R).

% Heuristic: misplaced tiles
heuristic(State, H) :-
    goal(Goal),
    count_misplaced(State, Goal, H).

count_misplaced([], [], 0).
count_misplaced([0|T1], [_|T2], H) :-
    count_misplaced(T1, T2, H).
count_misplaced([H|T1], [H|T2], C) :-
    count_misplaced(T1, T2, C).
count_misplaced([H1|T1], [H2|T2], C) :-
    H1 \= H2,
    count_misplaced(T1, T2, C1),
    C is C1 + 1.

% Evaluate children
evaluate_children([], []).
evaluate_children([State|Rest], [[H, State] | ScoredRest]) :-
    heuristic(State, H),
    evaluate_children(Rest, ScoredRest).

% Print state
print_state([A,B,C,D,E,F,G,H,I]) :-
    write([A,B,C]), nl,
    write([D,E,F]), nl,
    write([G,H,I]), nl.
```

OUTPUT-

```
?- consult("C:/Users/Admin/Desktop/Rutik/Hill_Climbing").
true.

?- solve([1,2,3,4,5,6,0,7,8]).
Current State:
[1,2,3]
[4,5,6]
[0,7,8]

Current State:
[1,2,3]
[4,5,6]
[7,0,8]

Goal Reached:
[1,2,3]
[4,5,6]
[7,8,0]
true
```



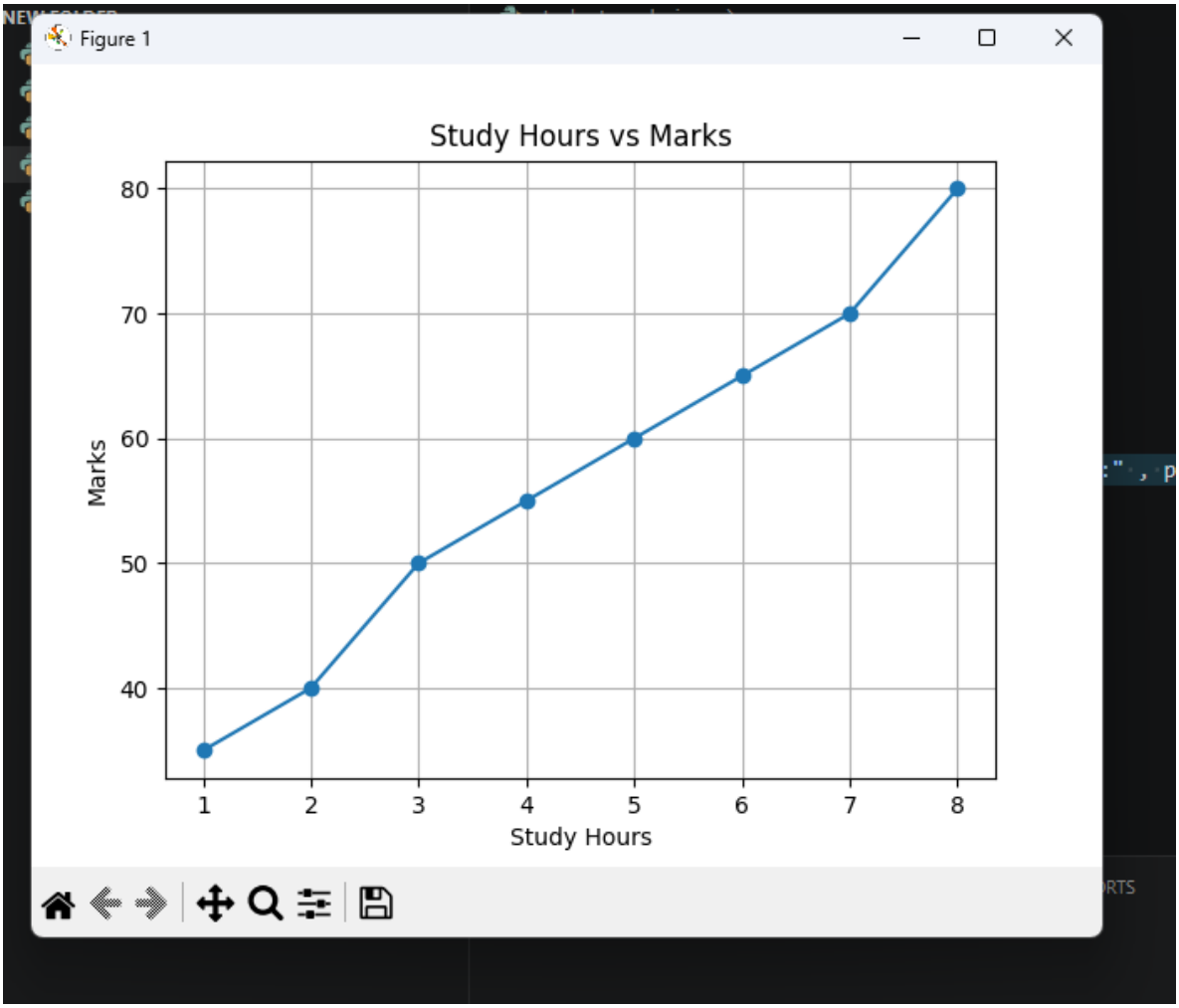
PRACTICAL NO 4
Introduction to Python Programming: Learn the different libraries - NumPy, Pandas, SciPy, Matplotlib, Scikit Learn.

CODE: student_analysis.py

```
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

hours = np.array([1,2,3,4,5,6,7,8])

marks = np.array([35,40,50,55,60,65,70,80])

data = pd.DataFrame({
    'Study_Hours': hours,
    'Marks' : marks
})

print("\nStudent Data:")
print(data)

mean_marks = np.mean(marks)
median_marks = np.median(marks)
mode_marks = stats.mode(marks)

print("\nStatistical Analysis")
print("Mean:", mean_marks)
print("Median:", median_marks)
print("Mode:", mode_marks)

plt.plot(hours, marks, marker='o')
plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.grid(True)

plt.show()

x = hours.reshape(-1,1)
y = marks
model = LinearRegression()

model.fit(x, y)

predicted_marks = model.predict([[9]])

print("\nMachihne Learning Prediction")
print("Predicted Marks for 9 Hours Study:" , predicted_marks[0])
```

OUTPUT:

```
Student Data:
   Study_Hours  Marks
0            1     35
1            2     40
2            3     50
3            4     55
4            5     60
5            6     65
6            7     70
7            8     80

Statistical Analysis
Mean: 56.875
Median: 57.5
Mode: ModeResult(mode=np.int64(35), count=np.int64(1))
```



PRACTICAL NO 5
Implement Perceptron algorithm for OR operation

CODE: Percepton operation.py

```
inputs = [(0,0),(0,1),(1,0),(1,1)]
labels = [0,1,1,1]

w1, w2=0.0,0.0
bias = 0.0
learning_rate = 0.1
epochs = 10

def step(x):
   return 1 if x >= 0 else 0

   for _ in range(epochs):
      for (x1,x2), label in zip(inputs, labels):
         linear_output = w1*x1 + w2*x2 + bias
         prediction = step(linear_output)

         error = label - prediction
         w1 += learning_rate * error * x1
         w2 += learning_rate * error * x2
         bias += learning_rate * KeyError

print("Trained weights:" , w1,w2)
print("Bias:", bias)

for x1, x2 in inputs:
   output = step(w1*x1 + w2*x2 +bias )
   print(f"{x1} OR {x2} = {output}")
```

OUTPUT:

PRACTICAL NO 6

Improve the prediction accuracy by estimating the weight values for the training data using stochastic gradient descent. (Perceptron)

CODE:

```
#Perceptron using Stochastic Gradient

import numpy as np
x=np.array([ [0,0],[0,1],[1,0],[1,1]
])

y=np.array([0,0,0,1])

weights=np.zeros(2)
bias=0

lr=0.1 epochs=10

def activation(z):
   return 1 if z>=0 else 0

for epoch in range(epochs):
   print(f"\nEpoch {epoch +1}")

   for i in range(len(x)):
      z=np.dot(x[i],weights)+bias
      y_pred=activation(z)
      error=y[i] - y_pred
      weights = weights + lr * error * x[i]
      bias = bias + lr * error

      print(f"Input: {x[i]}, Predicted: {y_pred}, Error: {error}")

print("\nFinal Weights:", weights)
print("Final Bias:", bias)

print("\nTesting Perceptron")
for i in range(len(x)):
   z = np.dot(x[i], weights) + bias
   y_pred = activation(z)
   print(f"Input: {x[i]} -> Output: {y_pred}")
```

OUTPUT:

```
PS C:\Users\admin\Desktop\AIML> python -u "c:\Users\admin\Desktop\AIML\perceptron using  Stochastic Gradient.py"

Epoch 1
Input: [0 0], Predicted: 1, Error: -1
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 0, Error: 1

Epoch 2
Input: [0 0], Predicted: 1, Error: -1
Input: [0 1], Predicted: 1, Error: -1
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 0, Error: 1

Epoch 3
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 1, Error: -1
Input: [1 0], Predicted: 1, Error: -1
Input: [1 1], Predicted: 0, Error: 1

Epoch 4
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0

Epoch 5
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0

Epoch 6
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0

Epoch 7
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0

Epoch 8
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0
```

```
Epoch 9
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0

Epoch 10
Input: [0 0], Predicted: 0, Error: 0
Input: [0 1], Predicted: 0, Error: 0
Input: [1 0], Predicted: 0, Error: 0
Input: [1 1], Predicted: 1, Error: 0

Final Weights: [0.2 0.1]
Final Bias: -0.20000000000000004

Testing Perceptron
Input: [0 0] -> Output: 0
Input: [0 1] -> Output: 0
Input: [1 0] -> Output: 0
Input: [1 1] -> Output: 1
```

PRACTICAL NO 7

Implement Adaline algorithm for AND operation

CODE:

```
#Perceptron using Stochastic Gradient

import numpy as np
x=np.array([ [0,0],[0,1],[1,0],[1,1]
])

y=np.array([0,0,0,1])

weights=np.zeros(2)
bias=0

lr=0.1 epochs=10

def activation(z):
    return 1 if z>=0 else 0

for epoch in range(epochs):
    print(f"\nEpoch {epoch +1}")

    for i in range(len(x)):
        z=np.dot(x[i],weights)+bias
        y_pred=activation(z)
        error=y[i] - y_pred
        weights = weights + lr * error * x[i]
        bias = bias + lr * error

        print(f"Input: {x[i]}, Predicted: {y_pred}, Error: {error}")

print("\nFinal Weights:", weights)
print("Final Bias:", bias)

print("\nTesting Perceptron")
for i in range(len(x)):
    z = np.dot(x[i], weights) + bias
    y_pred = activation(z)
    print(f"Input: {x[i]} -> Output: {y_pred}")
```

OUTPUT:

```
PS C:\Users\admin\Desktop\AIML> python -u "c:\Users\admin\Desktop\AIML\Adaline algorithm.py"
Initial Weights:  [0. 0.]
Initial Bias : 0

Epoch 1
Input : [0 0], Target : 0, Output : 0.00
Input : [0 1], Target : 0, Output : 0.00
Input : [1 0], Target : 0, Output : 0.00
Input : [1 1], Target : 1, Output : 0.00
Updated Weights : [0.1 0.1]
Updated Bias : 0.1

Epoch 2
Input : [0 0], Target : 0, Output : 0.10
Input : [0 1], Target : 0, Output : 0.19
Input : [1 0], Target : 0, Output : 0.17
Input : [1 1], Target : 1, Output : 0.22
Updated Weights : [0.16112 0.15922]
Updated Bias : 0.13

Epoch 3
Input : [0 0], Target : 0, Output : 0.13
Input : [0 1], Target : 0, Output : 0.28
Input : [1 0], Target : 0, Output : 0.25
Input : [1 1], Target : 1, Output : 0.33
Updated Weights : [0.20258054 0.19808926]
Updated Bias : 0.13

Epoch 4
Input : [0 0], Target : 0, Output : 0.13
Input : [0 1], Target : 0, Output : 0.32
Input : [1 0], Target : 0, Output : 0.29
Input : [1 1], Target : 1, Output : 0.40
Updated Weights : [0.23371999 0.22650395]
Updated Bias : 0.12

Epoch 5
Input : [0 0], Target : 0, Output : 0.12
Input : [0 1], Target : 0, Output : 0.33
Input : [1 0], Target : 0, Output : 0.31
Input : [1 1], Target : 1, Output : 0.44
Updated Weights : [0.25910386 0.24927607]
Updated Bias : 0.1

Epoch 6
Input : [0 0], Target : 0, Output : 0.10
Input : [0 1], Target : 0, Output : 0.34
Input : [1 0], Target : 0, Output : 0.31
Input : [1 1], Target : 1, Output : 0.47
Updated Weights : [0.28099036 0.26876266]
Updated Bias : 0.08
```

```
Epoch 7
Input : [0 0], Target : 0, Output : 0.08
Input : [0 1], Target : 0, Output : 0.34
Input : [1 0], Target : 0, Output : 0.32
Input : [1 1], Target : 1, Output : 0.49
Updated Weights : [0.3005204  0.28613437]
Updated Bias : 0.06

Epoch 8
Input : [0 0], Target : 0, Output : 0.06
Input : [0 1], Target : 0, Output : 0.34
Input : [1 0], Target : 0, Output : 0.32
Input : [1 1], Target : 1, Output : 0.51
Updated Weights : [0.31829198 0.30198762]
Updated Bias : 0.03

Epoch 9
Input : [0 0], Target : 0, Output : 0.03
Input : [0 1], Target : 0, Output : 0.33
Input : [1 0], Target : 0, Output : 0.32
Input : [1 1], Target : 1, Output : 0.52
Updated Weights : [0.33463769 0.31664003]
Updated Bias : 0.01

Epoch 10
Input : [0 0], Target : 0, Output : 0.01
Input : [0 1], Target : 0, Output : 0.33
Input : [1 0], Target : 0, Output : 0.31
Input : [1 1], Target : 1, Output : 0.53
Updated Weights : [0.34975884 0.3302731 ]
Updated Bias : -0.01

Testing Adaline for AND Operation
Input : [0 0] -> Output : 0
Input : [0 1] -> Output : 0
Input : [1 0] -> Output : 0
Input : [1 1] -> Output : 1
```

Machine Learning Section

PRACTICAL NO 1

Implementation of Features Extraction and Selection, Normalization, Transformation, Principal Components Analysis.

CODE:

```
import pandas as pd import numpy as np
from sklearn.preprocessing import StandardScaler, MinMaxScaler, PowerTransformer, LabelEncoder from sklearn.feature_selection import SelectKBest, f_classif, VarianceThreshold
from sklearn.decomposition import PCA



# Load the healthcare dataset file_path = "health_dataset.csv" df = pd.read_csv(file_path)

# Display first few rows
display(df.head())



# Identify non-numeric columns
categorical_columns = df.select_dtypes(include=['object']).columns print("Categorical Columns:", categorical_columns)

# Convert categorical columns using Label Encoding
label_encoders = {}
for col in categorical_columns: le = LabelEncoder()
df[col] = le.fit_transform(df[col]) label_encoders[col] = le

# Ensure all columns are numeric
df = df.apply(pd.to_numeric, errors='coerce') # Converts non-numeric values to NaN df.fillna(0, inplace=True) # Replace NaNs with 0

# Separate features and target
target_column = 'Disease Risk Score' if target_column not in df.columns:
raise ValueError("Target column not found in dataset")



X = df.drop(columns=[target_column]) y = df[target_column]

# Check for missing values in target and encode if necessary
if y.isnull().sum() > 0:
y.fillna(y.mode()[0], inplace=True) # Replace NaN with most frequent value
```

```
if y.dtype == 'object': le = LabelEncoder()
y = le.fit_transform(y)


# Remove constant features
var_thresh = VarianceThreshold(threshold=0) X = var_thresh.fit_transform(X)

# Feature Selection using ANOVA F-score
selector = SelectKBest(score_func=f_classif, k=min(8, X.shape[1])) # Ensure k does not exceed feature count
X_selected = selector.fit_transform(X, y) print(X_selected)
selected_features = [col for col, keep in zip(df.drop(columns=[target_column])
.columns,
selector.get_support()) if keep] print("Selected Features:", selected_features)

# Normalization using Min-Max Scaling
scaler = MinMaxScaler()
X_normalized = scaler.fit_transform(X_selected)


# Transformation using Power Transform (Box-Cox or Yeo-Johnson)
power_transformer = PowerTransformer(method='yeo-johnson') # Use 'box-cox' if no negative values
X_transformed = power_transformer.fit_transform(X_normalized)


# Dimensionality Reduction using PCA
pca_components = min(5, X_transformed.shape[1]) # Ensure PCA components do not exceed feature count
pca = PCA(n_components=pca_components) X_pca = pca.fit_transform(X_transformed)
print("Explained Variance Ratio:", pca.explained_variance_ratio_)


# Convert processed data back to DataFrame
processed_df = pd.DataFrame(X_pca, columns=[f'PC{i+1}'
for i in range(X_pca.shape[1])])
processed_df['Disease Risk Score'] = y.reset_index(drop=True)


# Save processed dataset
processed_df.to_csv("processed_health_dataset.csv", index=False) print("Processed dataset saved successfully.")
```

OUTPUT:

	ID	Age	Height (cm)	Weight (kg)	Blood Pressure (BP)	Cholesterol Level	Diabetes	Physical Activity (hours/week)	Smoking Habit	Disease Risk Score
0	1	56	159	77	122	High	1	1.00	0	59.00
1	2	69	185	77	157	Normal	0	0.53	1	48.00
2	3	46	163	93	122	Low	1	9.59	0	42.05
3	4	32	180	79	103	Normal	1	8.47	0	78.47
4	5	60	197	111	110	Normal	1	3.55	1	63.94

```
Categorical Columns: Index(['Cholesterol Level'], dtype='object')
[[ 56.   159.    77.   122.     0.     1.     1.     0.  ]
 [ 69.   185.    77.   157.     2.     0.     0.53   1.  ]
 [ 46.   163.    93.   122.     1.     1.     9.59   0.  ]
 [ 32.   180.    79.   103.     2.     1.     8.47   0.  ]
 [ 60.   197.   111.   110.     2.     1.     3.55   1.  ]
 [ 25.   164.   111.   137.     2.     0.     9.57   1.  ]
 [ 78.   157.    50.   109.     2.     1.     6.77   0.  ]
 [ 38.   163.    76.    97.     1.     1.     4.83   1.  ]
 [ 56.   172.   111.    96.     2.     1.     4.93   0.  ]
 [ 75.   189.    52.   156.     2.     1.     0.83   0.  ]
 [ 36.   170.   119.   106.     0.     0.     0.92   0.  ]
 [ 40.   165.    76.   122.     2.     1.     6.02   1.  ]
 [ 28.   194.    58.   137.     0.     0.     5.54   1.  ]
 [ 28.   167.   111.   165.     1.     0.     2.13   1.  ]
 [ 41.   196.    86.   148.     1.     0.     9.46   1.  ]
 [ 70.   173.   100.   175.     1.     0.     7.81   1.  ]
 [ 53.   175.    93.   111.     1.     0.     1.13   0.  ]
 [ 57.   174.    73.   119.     0.     0.     9.31   0.  ]
 [ 41.   194.   108.   127.     2.     1.     9.74   0.  ]
 [ 20.   190.    81.   140.     1.     0.     9.96   0.  ]
 [ 39.   178.   101.   143.     2.     0.     0.56   1.  ]
 [ 70.   164.   111.    97.     0.     0.     7.37   0.  ]
 [ 19.   194.   107.   116.     0.     1.     5.46   1.  ]
 [ 41.   150.   101.   116.     0.     1.     7.06   0.  ]
 [ 61.   174.    61.   110.     1.     0.     9.69   1.  ]
 [ 47.   156.    88.   119.     2.     1.     6.88   1.  ]
 [ 55.   158.    51.   117.     2.     0.     8.37   0.  ]
 [ 19.   173.    52.   153.     2.     0.     8.67   0.  ]
 [ 77.   150.   105.   158.     0.     1.     8.38   0.  ]
 [ 38.   193.   108.   150.     1.     0.     4.26   0.  ]
 [ 50.   157.    51.   137.     0.     1.     2.23   1.  ]
 [ 29.   173.    51.   108.     0.     1.     3.97   0.  ]
 [ 75.   160.   103.    93.     2.     0.     8.92   0.  ]
 [ 39.   166.    50.   124.     0.     0.     1.47   1.  ]
 [ 78.   157.    68.   153.     2.     1.     5.13   0.  ]
 [ 61.   184.    51.   138.     1.     0.     2.33   0.  ]
```

```
[ 42.   184.   102.   106.     2.     0.     5.81   0.  ]
[ 66.   182.    93.   133.     1.     0.     8.63   0.  ]
[ 44.   154.    81.   119.     2.     1.     8.8    0.  ]
[ 76.   191.   119.   135.     0.     1.     2.37   1.  ]
[ 59.   188.    81.    95.     0.     0.     9.08   0.  ]
[ 45.   190.   117.   126.     1.     1.     5.92   0.  ]
[ 77.   177.   104.   113.     1.     1.     3.5    0.  ]
[ 33.   156.   105.   135.     1.     1.     7.08   0.  ]
[ 32.   158.    66.   142.     2.     1.     4.82   0.  ]
[ 79.   157.    87.   149.     2.     1.     3.78   1.  ]
[ 79.   161.    73.   152.     1.     0.     7.05   0.  ]
[ 64.   183.   118.   174.     0.     0.     2.49   1.  ]
[ 79.   182.   119.   121.     2.     0.     3.3    1.  ]
[ 68.   197.    60.   176.     2.     1.     4.34   0.  ]
[ 61.   172.    65.   122.     2.     1.     2.54   0.  ]
[ 72.   173.   108.   156.     1.     0.     4.05   0.  ]
[ 69.   186.   119.   107.     1.     1.     5.71   0.  ]
[ 74.   184.    52.   114.     1.     0.     7.41   1.  ]
[ 20.   193.    69.   143.     1.     0.     7.67   0.  ]
[ 54.   189.   108.   147.     1.     0.     8.23   0.  ]
[ 68.   171.    85.   156.     0.     1.     7.44   0.  ]
[ 24.   176.    68.   135.     0.     1.     6.81   0.  ]
[ 38.   184.   116.   113.     1.     0.     2.38   0.  ]
[ 26.   150.    68.   121.     2.     0.     4.     0.  ]
[ 56.   184.    69.   136.     1.     0.     4.78   1.  ]
[ 35.   186.   101.   175.     0.     0.     0.83   1.  ]
[ 21.   196.    82.   112.     0.     1.     5.28   0.  ]
[ 42.   163.    89.   155.     2.     1.     4.36   0.  ]
[ 77.   152.    88.   116.     0.     0.     8.02   0.  ]
[ 31.   150.    50.    91.     1.     0.     9.78   1.  ]
[ 67.   154.    60.   179.     2.     1.     5.56   0.  ]
[ 75.   175.   106.   106.     1.     1.     3.23   1.  ]
[ 26.   163.    99.   122.     1.     0.     0.43   0.  ]
[ 43.   188.    72.    98.     2.     0.     9.25   0.  ]
[ 70.   176.    80.   132.     1.     0.     9.19   0.  ]
[ 19.   158.    91.   137.     0.     1.     2.53   0.  ]
[ 37.   164.    56.   128.     1.     1.     6.95   1.  ]
[ 45.   164.    65.   131.     2.     1.     0.75   1.  ]
[ 64.   175.   109.   115.     2.     0.     1.66   0.  ]
[ 77.   191.    51.   139.     2.     0.     2.17   0.  ]
[ 24.   162.    50.   114.     1.     1.     2.94   1.  ]
[ 61.   181.    97.   113.     2.     1.     9.96   1.  ]
[ 78.   188.    61.   102.     2.     0.     6.97   1.  ]
[ 25.   198.   118.   149.     1.     1.     3.84   1.  ]
[ 64.   181.    86.    96.     0.     1.     7.37   0.  ]
[ 52.   153.    81.   146.     1.     0.     9.15   1.  ]
[ 31.   179.    58.   125.     2.     1.     9.59   0.  ]
[ 34.   186.    68.   134.     2.     0.     0.58   1.  ]
[ 53.   172.    97.   109.     1.     0.     3.95   1.  ]
```

```
 [ 67.   188.    52.   154.     0.     0.     1.07   1.  ]
 [ 57.   194.    69.    97.     2.     0.     3.36   1.  ]
 [ 21.   164.    73.   105.     2.     0.     1.7    1.  ]
 [ 19.   192.   103.   103.     2.     0.     6.47   0.  ]
 [ 79.   178.    82.   165.     1.     0.     3.88   0.  ]
 [ 23.   185.    73.   176.     2.     1.     2.29   1.  ]
 [ 71.   162.    85.   104.     0.     0.     2.66   1.  ]
 [ 59.   181.    87.   155.     2.     0.     3.6    0.  ]
 [ 21.   156.    74.   121.     2.     1.     2.6    1.  ]
 [ 71.   171.    67.   176.     0.     1.     4.53   1.  ]
 [ 46.   177.   115.   152.     2.     1.     0.32   0.  ]
 [ 35.   151.   103.   175.     1.     0.     2.8    1.  ]
 [ 43.   191.    84.   140.     1.     1.     4.11   1.  ]
 [ 61.   194.   110.   114.     1.     0.     6.03   0.  ]
 [ 51.   155.    90.   147.     0.     0.     2.71   0.  ]]
Selected Features: ['Age', 'Height (cm)', 'Weight (kg)', 'Blood Pressure (BP)', 'Cholesterol Level', 'Diabetes', 'Physical Activity (hours/week)', 'Smo
king Habit']
Explained Variance Ratio:
[0.17300838 0.15542009 0.13356947 0.12778434 0.12436891]
Processed dataset saved successfully.
```

PRACTICAL NO 2

Implementation of Logistic regression

CODE:

```
import numpy as np
from sklearn import datasets iris = datasets.load_iris() print(type(iris))
print(list(iris.keys()))
X = iris["data"][:,3:] # petal width
y = (iris["target"] == 2).astype(np.int64) # 1 if Iris-Virginica, else 0
from sklearn.linear_model import LogisticRegression
log_reg = LogisticRegression(solver="lbfgs", random_state=42) log_reg.fit(X,y)
import matplotlib.pyplot as plt
X_new = np.linspace(0,3,1000).reshape(-1,1) y_proba = log_reg.predict_proba(X_new)
plt.plot(X_new, y_proba[:,1],"g-")
plt.plot(X_new, y_proba[:,0], "b--") plt.xlabel('Petal width (cm)') plt.ylabel('Probability')
plt.legend(['Prob(Iris-Virginica)','Prob(Not Iris-Virginica)']) print(log_reg.predict(X))
print(log_reg.coef_) print(log_reg.intercept_)
```

2. Implementation of Logistic regression (two features)

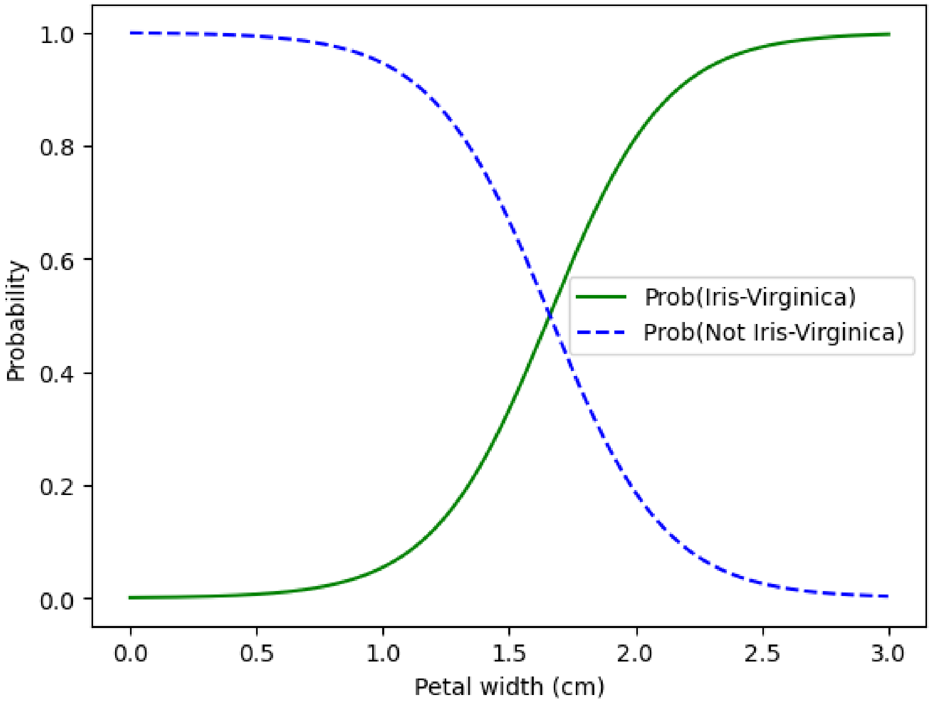
#Problem Statement 2: Logistic Regression for predicting class using two features: Petal length and width.

CODE:

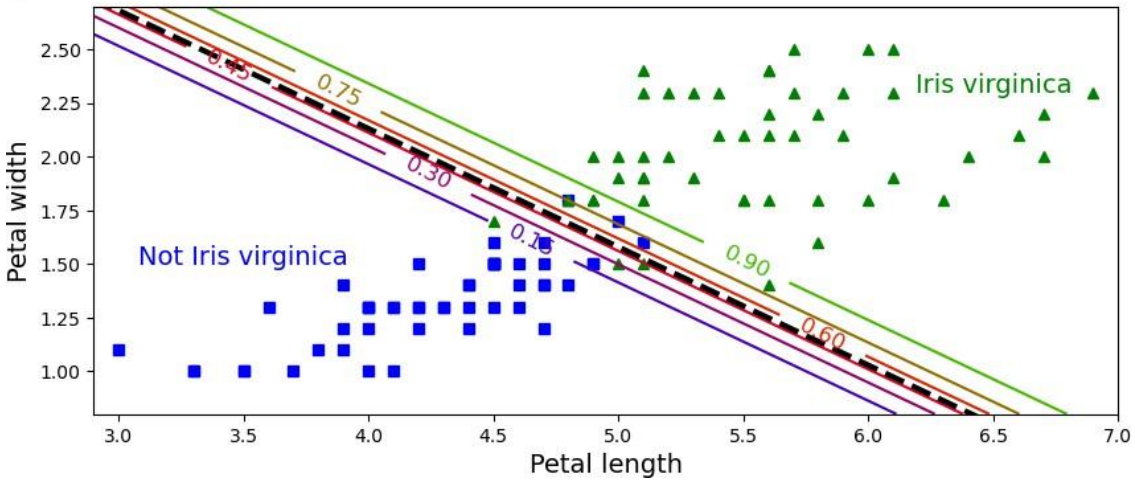
```
from sklearn.linear_model import LogisticRegression X = iris["data"][:, (2, 3)] # petal
length, petal width y = (iris["target"] == 2).astype(np.int64)
log_reg2 = LogisticRegression(solver="lbfgs", C=10**10, random_state=42)
log_reg2.fit(X, y)
x0, x1 = np.meshgrid(
np.linspace(2.9, 7, 500).reshape(-1, 1),
np.linspace(0.8, 2.7, 200).reshape(-1, 1),
)
X_new = np.c_[x0.ravel(), x1.ravel()] print(X_new.shape)
y_proba = log_reg2.predict_proba(X_new) plt.figure(figsize=(10, 4))
plt.plot(X[y==0, 0], X[y==0, 1], "bs")
plt.plot(X[y==1, 0], X[y==1, 1], "g^") zz = y_proba[:, 1].reshape(x0.shape)
contour = plt.contour(x0, x1, zz, cmap=plt.cm.brg) left_right = np.array([2.9, 7])
boundary = -(log_reg2.coef_[0][0] * left_right + log_reg2.intercept_[0]) /
log_reg2.coef_[0][1] plt.clabel(contour, inline=1, fontsize=12)
plt.plot(left_right, boundary, "k--", linewidth=3)
plt.text(3.5, 1.5, "Not Iris virginica", fontsize=14, color="b", ha="center") plt.text(6.5, 2.3,
"Iris virginica", fontsize=14, color="g", ha="center") plt.xlabel("Petal length", fontsize=14)
plt.ylabel("Petal width", fontsize=14) plt.axis([2.9, 7, 0.8, 2.7])
```

OUTPUT:

```
<class 'sklearn.utils._bunch.Bunch'>
['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename', 'data_module']
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0
 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1 1 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1]
[[4.33251974]]
[-7.19342106]
```



```
(100000, 2)
(np.float64(2.9), np.float64(7.0), np.float64(0.8), np.float64(2.7))
```



PRACTICAL NO 3

Implementation of Classifying data using Support Vector Machine (SVM)- Linear and Non-Linear SVM Classification

1. Linear SVM

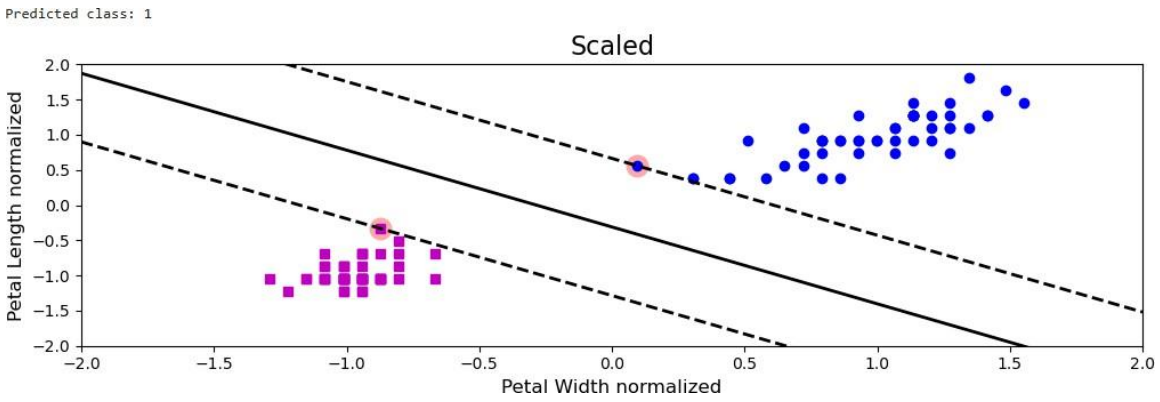
```
from sklearn.svm import SVC from sklearn import datasets import numpy as np
# Load Iris dataset
iris = datasets.load_iris()
X = iris["data"][:, (2, 3)] # Select petal length and petal width
y = iris["target"]
# Select only Setosa and Versicolor classes
setosa_or_versicolor = (y == 0) | (y == 1) X = X[setosa_or_versicolor]
y = y[setosa_or_versicolor]


# SVM Classifier model with a large but finite C value
svm_clf = SVC(kernel="linear", C=1e10) # Large C approximates a hard margin
svm_clf.fit(X, y)


# Make a prediction
prediction = svm_clf.predict([[2.4, 3.1]]) print("Predicted class:", prediction[0])
import matplotlib
import matplotlib.pyplot as plt
def plot_svc_decision_boundary(svm_clf, xmin, xmax): w = svm_clf.coef_[0]
b = svm_clf.intercept_[0]
# At the decision boundary, w0*x0 + w1*x1 + b = 0 # => x1 = -w0/w1 * x0 - b/w1
x0 = np.linspace(xmin, xmax, 200)

decision_boundary = -w[0]/w[1] * x0 - b/w[1] margin = 1/w[1]
gutter_up = decision_boundary + margin gutter_down = decision_boundary - margin svs =
svm_clf.support_vectors_
plt.scatter(svs[:, 0], svs[:, 1], s=180, facecolors='#FFAAAA') plt.plot(x0, decision_boundary, "k-",
linewidth=2) plt.plot(x0, gutter_up, "k--", linewidth=2)
plt.plot(x0, gutter_down, "k--", linewidth=2)
#plot the decision boundaries import numpy as np plt.figure(figsize=(12,3.2))
from sklearn.preprocessing import StandardScaler scaler = StandardScaler()
X_scaled = scaler.fit_transform(X) svm_clf.fit(X_scaled, y)
plt.plot(X_scaled[:, 0][y==1], X_scaled[:, 1][y==1], "bo")
plt.plot(X_scaled[:, 0][y==0], X_scaled[:, 1][y==0], "ms") plot_svc_decision_boundary(svm_clf, -2,
2)
plt.xlabel("Petal Width normalized", fontsize=12) plt.ylabel("Petal Length normalized", fontsize=12)
plt.title("Scaled", fontsize=16)
plt.axis([-2, 2, -2, 2])
```

2. Non-Linear SVM Code:

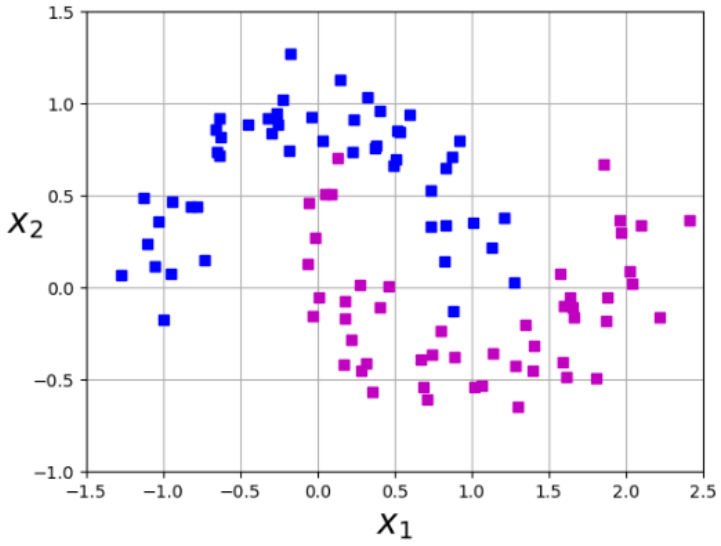
```
from sklearn.datasets import make_moons from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures from sklearn.preprocessing import
StandardScaler
from sklearn.svm import SVC import numpy as np
```

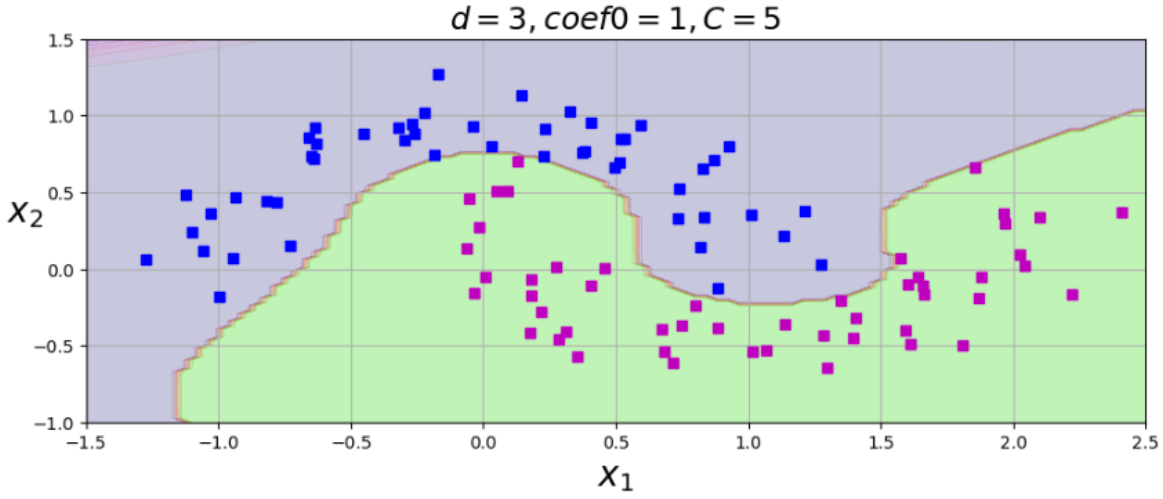
```
%matplotlib inline import matplotlib
import matplotlib.pyplot as plt
X, y = make_moons(n_samples=100, noise=0.15, random_state=42) print(X.shape)
print(y.shape)
#define a function to plot the dataset def plot_dataset(X, y, axes):
plt.plot(X[:, 0][y==0], X[:, 1][y==0], "bs")
plt.plot(X[:, 0][y==1], X[:, 1][y==1], "ms")
plt.axis(axes)
plt.grid(True, which='both') plt.xlabel(r"$x_1$", fontsize=20)
plt.ylabel(r"$x_2$", fontsize=20, rotation=0) #Let's have a look at the data we have generated
plot_dataset(X, y, [-1.5, 2.5, -1, 1.5])
#C controls the width of the street #Degree of data
#create a pipeline to create features, scale data and fit the model
polynomial_svm_clf = Pipeline((
("poly_features", PolynomialFeatures(degree=3)), ("scalar", StandardScaler()),
("svm_clf", SVC(kernel="poly", degree=10, coef0=1, C=5))
))
#call the pipeline
polynomial_svm_clf.fit(X,y)
#define a function plot the decision boundaries
def plot_predictions(clf, axes):
#create data in continous linear space x0s = np.linspace(axes[0], axes[1], 100) x1s =
np.linspace(axes[2], axes[3], 100) x0, x1 = np.meshgrid(x0s, x1s)
X = np.c_[x0.ravel(), x1.ravel()]
y_pred = clf.predict(X).reshape(x0.shape)
y_decision = clf.decision_function(X).reshape(x0.shape) plt.contourf(x0, x1, y_pred,
cmap=plt.cm.brg, alpha=0.2)
plt.contourf(x0, x1, y_decision, cmap=plt.cm.brg, alpha=0.1)
#plot the decision boundaries
plt.figure(figsize=(11, 4))
#plot the dataset
plot_dataset(X, y, [-1.5, 2.5, -1, 1.5])
#plot the decision boundaries
plot_predictions(polynomial_svm_clf, [-1.5, 2.5, -1, 1.5]) plt.title(r"$d=3, coef0=1, C=5$",
fontsize=18)
```

OUTPUT:



```
(100, 2)
(100,)
```


1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
x_2
x_1


$d = 3, coef0 = 1, C = 5$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
x_2
x_1

PRACTICAL NO 4

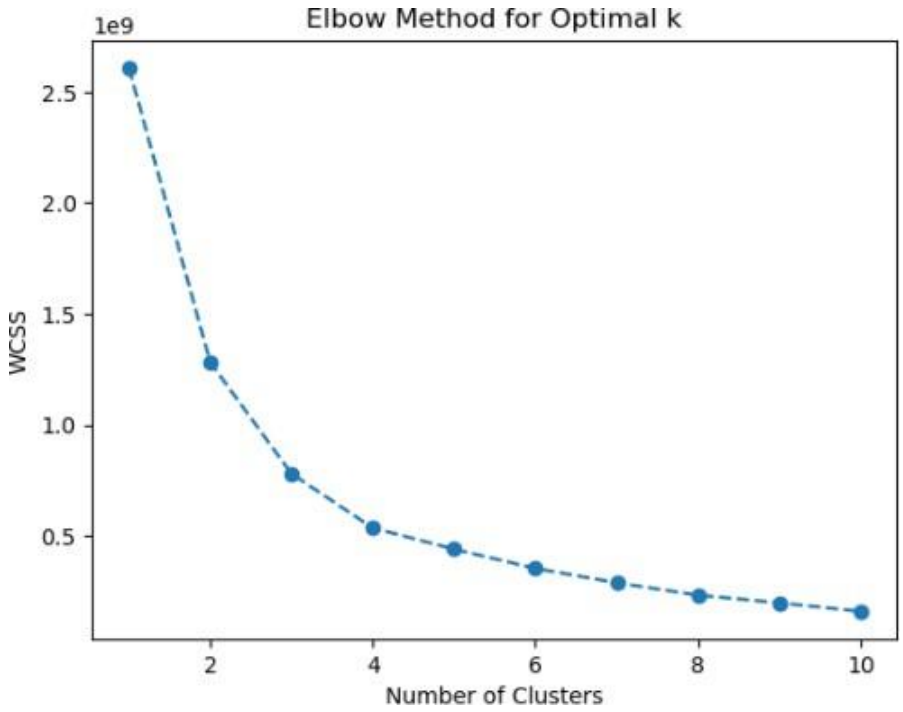
Implementation of Bagging Algorithm: Random Forest

CODE:

```
import pandas as pd import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
# Load the dataset
df = pd.read_csv("clustering.csv")
# Display first few rows of the dataset
print(df.head())
# Drop missing values
df_cleaned = df.dropna()
# Selecting numerical columns for clustering
numerical_cols = df_cleaned.select_dtypes(include=[np.number]).columns print("Numerical columns used for clustering:", numerical_cols.tolist())
# Feature selection for clustering (Modify as needed)
X = df_cleaned[numerical_cols]
# Apply the Elbow Method
wcss = [] # Within-cluster sum of squares
for i in range(1, 11): # Trying different cluster numbers from 1 to 10
kmeans = KMeans(n_clusters=i, random_state=42, n_init=10) kmeans.fit(X)
wcss.append(kmeans.inertia_)
# Plot the Elbow Method
plt.plot(range(1, 11), wcss, marker='o', linestyle='--') plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method for Optimal k') plt.show()
# Choose optimal k (Modify based on the elbow plot observation)

k_optimal = 3 # Example choice, change based on your dataset # Apply K-Means with the optimal number of clusters
kmeans = KMeans(n_clusters=k_optimal, random_state=42, n_init=10) df_cleaned['Cluster'] = kmeans.fit_predict(X)
# Display clustered data
print(df_cleaned.head())
# Plot the clusters (for 2D visualization, choose two relevant features)
plt.scatter(df_cleaned[numerical_cols[0]], df_cleaned[numerical_cols[1]], c=df_cleaned['Cluster'], cmap='viridis')
plt.xlabel(numerical_cols[0])  plt.ylabel(numerical_cols[1])
plt.title(f'K-Means Clustering (k={k_optimal})') plt.colorbar(label='Cluster')
plt.show()
```

OUTPUT:

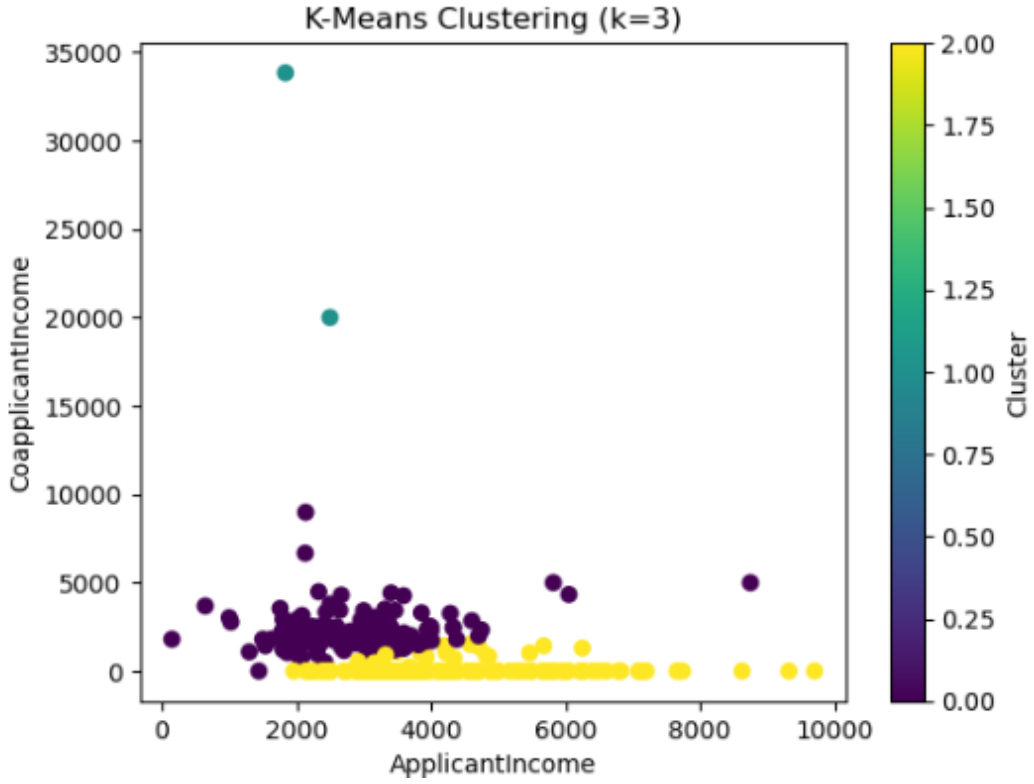
```
    Loan_ID Gender Married Dependents     Education Self_Employed  \
0  LP001003   Male     Yes          1      Graduate            No
1  LP001005   Male     Yes          0      Graduate           Yes
2  LP001006   Male     Yes          0  Not Graduate            No
3  LP001008   Male      No          0      Graduate            No
4  LP001013   Male     Yes          0  Not Graduate            No

   ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0             4583             1508.0       128.0             360.0
1             3000                0.0        66.0             360.0
2             2583             2358.0       120.0             360.0
3             6000                0.0       141.0             360.0
4             2333             1516.0        95.0             360.0

   Credit_History Property_Area Loan_Status
0             1.0         Rural           N
1             1.0         Urban           Y
2             1.0         Urban           Y
3             1.0         Urban           Y
4             1.0         Urban           Y
Numerical columns used for clustering: ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount', 'Loan_Amount_Term', 'Credit_History']
```



	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	\
0	LP001003	Male	Yes	1	Graduate	No	
1	LP001005	Male	Yes	0	Graduate	Yes	
2	LP001006	Male	Yes	0	Not Graduate	No	
3	LP001008	Male	No	0	Graduate	No	
4	LP001013	Male	Yes	0	Not Graduate	No	

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	\
0	4583	1508.0	128.0	360.0	
1	3000	0.0	66.0	360.0	
2	2583	2358.0	120.0	360.0	
3	6000	0.0	141.0	360.0	
4	2333	1516.0	95.0	360.0	

	Credit_History	Property_Area	Loan_Status	Cluster
0	1.0	Rural	N	2
1	1.0	Urban	Y	2
2	1.0	Urban	Y	0
3	1.0	Urban	Y	2
4	1.0	Urban	Y	0



PRACTICAL NO 5
Implementation of Bagging Algorithm: Random Forest

CODE:

```
import numpy as np import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score,confusion_matrix from sklearn.decomposition import PCA
# Load dataset
iris = load_iris()
X = iris.data y = iris.target
# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Initialize and train the Random Forest Classifier
rf_classifier = RandomForestClassifier(n_estimators=100, random_state=42) rf_classifier.fit(X_train,
y_train)

# Make predictions
y_pred = rf_classifier.predict(X_test)


# Evaluate the model
cm=confusion_matrix(y_test, y_pred)
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy of Random Forest Classifier: {accuracy * 100:.2f}%') print("Confusion Matrix:\n", cm)
```

OUTPUT:

```
Accuracy of Random Forest Classifier: 100.00%
Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

PRACTICAL NO 6

Implementation of Boosting Algorithms: AdaBoost, Stochastic Gradient Boosting, Voting Ensemble

CODE:

```
import numpy as np import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier,
GradientBoostingClassifier, VotingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA
# Load dataset
iris = load_iris()
X = iris.data y = iris.target

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# Random Forest Classifier
rf_classifier = RandomForestClassifier(n_estimators=100, random_state=42) rf_classifier.fit(X_train,
y_train)
y_pred_rf = rf_classifier.predict(X_test)
accuracy_rf = accuracy_score(y_test, y_pred_rf)
print(f'Accuracy of Random Forest Classifier: {accuracy_rf * 100:.2f}%')


# AdaBoost Classifier
adaboost = AdaBoostClassifier(base_estimator=DecisionTreeClassifier(max_depth=1),
n_estimators=50, random_state=42)

adaboost.fit(X_train, y_train)
y_pred_adaboost = adaboost.predict(X_test)
accuracy_adaboost = accuracy_score(y_test, y_pred_adaboost)
print(f'Accuracy of AdaBoost Classifier: {accuracy_adaboost * 100:.2f}%')


# Gradient Boosting Classifier
gb_classifier = GradientBoostingClassifier(n_estimators=100,
```

```
learning_rate=0.1, random_state=42, subsample=0.8) gb_classifier.fit(X_train, y_train)
y_pred_gb = gb_classifier.predict(X_test)
accuracy_gb = accuracy_score(y_test, y_pred_gb)
print(f'Accuracy of Gradient Boosting Classifier: {accuracy_gb * 100:.2f}%')


# Voting Classifier (Ensemble of Logistic Regression, Decision Tree, and Random Forest)
voting_classifier = VotingClassifier(estimators=[ ('lr', LogisticRegression()),
('dt', DecisionTreeClassifier()),
('rf', RandomForestClassifier(n_estimators=100))
], voting='hard')
voting_classifier.fit(X_train, y_train)
y_pred_voting = voting_classifier.predict(X_test)
accuracy_voting = accuracy_score(y_test, y_pred_voting)
print(f'Accuracy of Voting Classifier: {accuracy_voting * 100:.2f}%')


# Reduce dimensions for visualization pca = PCA(n_components=2) X_reduced =
pca.fit_transform(X)

# Scatter plot of the dataset
plt.figure(figsize=(8, 6))
plt.scatter(X_reduced[:, 0], X_reduced[:, 1], c=y, cmap='viridis', edgecolor='k', alpha=0.7)

plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.title('Iris Dataset Visualization with PCA') plt.colorbar(label='Class Labels')
plt.show()
```

OUTPUT:

```
Accuracy of Random Forest Classifier: 100.00%
Accuracy of AdaBoost Classifier: 93.33%
Accuracy of Gradient Boosting Classifier: 100.00%
Accuracy of Voting Classifier: 100.00%
```